Corrected and conduct of chapter 9

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. $2^7 = 128$. *Action:* Reconstitute the product by seven factors.
2. $\frac{1}{9}$. *Action:* Back to the reverse power.
3. $x = 2$. *Action:* Isolate the unknown while maintaining equivalence.
4. $2x + 2$. *Action:* Apply the sum of the derivatives.
5. $0{,}8$. *Action:* Subtract the rate to 1.
6. 5×2^n. *Action:* Count the n multiplications.

Research activity

The condition $f\prime = f$ Motivation$y \approx hy$. With h= $\frac{1}{4}$, the values are 1 ; 1.25 ; 1.5625 ; 1.953125.
construction approximates the function and does not replace its characterization.

Immediate applications

1. $y = e + e(x - 1) = ex$.
2. e^{2x+1} ; $x + 2 > 2x$, therefore $x < 2$.
3. $g\prime(t) = -6e^{-2t} < 0$ on $\mathbb{R}$: Strictly decreasing.
4. $u_n = 5(e^{0{,}3})^n$; the report on two units is $e^{0{,}6}$.
5. Apply demonstrated inequality with argument 2 x ; equality only in x=0.

Exercises: approach and results

Exercise 9.1 · Transform. e^2 ; e^{5x} ; $1 + e^{-x}$. For the last expression, distribute and simplify products.

Exercise 9.2 · Equations and inequities. By strict growth: 2 $x - 1 = x$ + 3, i.e. x=4.
Then $-3x \geq x - 2$, or $x \leq \frac{1}{2}$.

Exercise 9.3 · Product and maximum. $f\prime = (1 - x)e^{-x}$; growth up to 1 and then decline.
Maximum $f(1) = \frac{1}{e}$. Values at bounds 0 and 4 e^{-4} lower.

Exercise 9.4 · A modeled population. 200 ; factor $e^{0{,}1}$; rate $e^{0{,}1} - 1 \approx 10{,}52$ per cent
resources and the capacity of the environment are not taken into account; it is a model over a period of time
The two curves requested go through (0;1); the first one grows, the second one decreases.
are symmetrical in relation to the vertical axis; see the figure in the section From Discrete to Continuous.

Exercise 9.5 · Search for a threshold. Test successively n=0,... with a limit of iterations. $Q(4) \approx 10{,}8268 \geq 10$, $Q(5) \approx 6{,}5668 < 10$; Demonstrated decay, therefore first integer 5.

Exercise 9.6 · Symmetry. $f(-x) = f(x)$: pair. $f\prime = e^{x} - e^{-x}$ has the sign of x, so minimum 2 in 0. Other evidence: $f(x) - 2 = (e^{\frac{x}{2}} - e^{-\frac{x}{2}})^{2}$.

Exercise 9.7 · Singleness under one hypothesis. $g\prime = f\prime e^{-x} - fe^{-x} = 0$. $\mathbb{R}$, g is constant, and g(0)=1. $f(x) = e^{x}$. This proof of uniqueness uses an exponential already constructed, it does not does not demonstrate existence.

Exercise 9.8 · Deferred review. $\frac{u_{n+1}}{u_n} = e^{2}$; first term e. Sum $e(1+e^{2}+e^{4}) = \frac{e(1-e^{6})}{1-e^{2}}$. The direct sum is also effective here.

CQFD, exercise 18 p. 187, questions a, b and c. a : e^{3x}. b : e^{2}, since e^{x+2} is divided by e^{x}, Always non-zero. c: e^{x+1}. Do not confuse the number e with the variable x.

CQFD, exercise 30 p. 187, questions a, b and c. a : $x \leq 0$. b : $2x > 1$, therefore $x > \frac{1}{2}$. c : $-3x \geq 2x + 1$, therefore $x \leq -\frac{1}{5}$. Strict growth of the exponential keeps meaning; division by -5 reverses it in c.

MCQ: answers and reasons

MCQ 1.

1. B. Initial condition.

2. A. Property of addition.

3. D. Factor -2.

4. Strict growth and $e^{0} = 1$.

5. B. Positiveness of the exponential.

MCQ 2.

1. C. Compare exhibitors.

2. A. Report of consecutive terms.

3. D. Value 1 and slope 1.

4. B. Product rule.

5. C. Minimum difference.

True or false

1. False: in 0, 1 is different from 2.

2. True: its derivative $-3e^{-3t}$ is negative.

3. False: the discreet rate is valid $e^{0,1} - 1$, about 10.52 per cent.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.